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Description - Contains theory for the TYME II Board.

1- TYME II BOARD

The TYME II (Tardis by YMS and Milwaukee Engineering) Board interfaces the TPS/ISE chassis to the rest of the system. It provides fiber optic transmitters/receivers for communication with the PAC, SRI, and MDS link. It transmits serial gradient data and clock waveforms optically to the gradient amplifiers. Electrical signals at RS-422 levels are driven from the TYME II board to control the RF Cabinet unblank and envelope feedback, and link the TPS/ISE to the operator's console. Buffer circuits are provided to accept an external trigger input, an AC line synchronization input and an output of four auxiliary analog channels. Lastly, the position of the TYME II board at one end of the backplane necessitates the inclusion of VME bus and other backplane terminations.

Serial data to/from the PAC and SRI are routed from the IPG board across the backplane on the J2 connector, where they are connected to the TYME II board. Test points are provided for service troubleshooting. These signals first pass through a multiplexor where it is possible to loop back IPG transmitted data for diagnostic purposes. Also serial data to/from MDS link is multiplexed at this location.

Signals at the RFBUS, comprise two sets of unblank and envelope feedback control signals for the amplifier in the RF cabinet. These signals are driven out as RS-422 differential signals. Two RS-422 drivers also buffer DAB and SSP bus signals to RS-422 level for future use.

A synchronization signal is generated from the AC line, and driven onto the back plane for use by the IPG board for gating purposes.

Serial physiological data from the IPG board is grouped as a byte with an eight bit shift register. Once assembled, each byte is latched. The latch holds the data which is available to a quadruple D to A converter until the next byte is available at the shift register. During this time data is written. Outputs are level shifted, buffered and driven to the outside world.

Auxiliary Analog Outputs provide the means for the gating signal processor on the IPG board to output physiological waveforms in a more conventional manner. These outputs are intended to drive external waveform displays, strip chart recorders, etc.

A Fiber Optic Repeater has been placed in the Penetration Panel to ensure the integrity of the optical signal communication between the TYME II board and the SRI and PAC in the Scan Room. The repeater board is a collection of optical Receivers and Transmitters that provide up to 10 db increase in the optical signal power. A MUX circuit provides a path for loopback test (except for SRI Reset signal) controlled by a jumper.

In addition, the TYME II provides a means for chassis voltage monitoring, thermal protection, SCB functionality, a VME interface and a Waveform Accelerator Module (WAM) interface.

1-1 Voltage Monitoring

Power supply levels are monitored by the TYME II board. Eight-bit digitized power supply levels are made available on the backplane via the VME interface. All the following chassis supply voltages are monitored: +5, -5.2, +12, -12 on the backplane; +5, +12 from the SCSI supply; +5, -5.2, +15, -15 from the CERD linear supplies; and +18, -18 from the Spectro supply. The monitoring software operates a mode which will scan all of the above voltages and creates a software interrupt if any of the supplies are out of tolerance.

1-2 Thermal protection

The TYME II provides temperature monitoring of the ISE chassis. Monitoring functions include a PDU shutdown if limits are exceeded. TYME II software is capable of providing a shutdown warning based on the status of the thermal monitor register. A temperature warning buzzer is incorporated into the TNF assembly. The Temperature Sensor trips at 40 degrees C. This trip shall provide a warning message to the system. The Temp Sensor also trips at 50 degrees C. This trip will signal the PDU to shut off power to all critical cabinets.

1-3 SCB Functionality

The Signal Collection Board (SCB) board is now incorporated into the TYME II which performs many of the same functions. The VME r/w registers provide the SCB functions, which are:

- Bore Fan On
- Alignment Lights On
- Bore Lights On
- Overnight Mode On
- 1 Spare Output

1-4 WAM INTERFACE

The TYME II supports the Waveform Accelerator Module (WAM) by providing an interface into the gradient data pipeline. The TYME II detects the presence of the WAM and re-routes B0 gradient data to the WAM. If the WAM is not present, the TYME II shall merely pass the B0 data through.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 2, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.